Points 1.5/18
(8.33%)

Aufgabe 2

Let $p \in \mathbb{Z}: p > 0$ be prime and let $K = \mathbb{Q}(\zeta)$ with $\zeta = \exp(\frac{2\pi i}{p})$. Let $l \neq p$ another prime in $\mathbb{Z}$, $l > 0$ and let $Q \subseteq \mathbb{Z}_K$ be prime ideal divider of $l\mathbb{Z}_K$

i)

t.s. $\zeta^a - 1 \notin Q$ if $a \in \mathbb{Z}$ coprime to p.

proof Since a, p coprime, we get $\zeta^a - 1 \neq 0$. ok

We assume $\zeta^a - 1 \in Q$. Then $(\zeta^a - 1)\mathbb{Z}_K \subseteq Q$. ok

By assumption there exists an ideal $I \subseteq \mathbb{Z}_K$ s.t.

$l\mathbb{Z}_K = Q \cdot I$. $\Rightarrow l\mathbb{Z}_K \subseteq Q$. ok

If $\zeta^a - 1 \in l\mathbb{Z}_K$, Th since K monogenic, it holds that $\mathbb{Z}_K = \mathbb{Z}[\zeta]$ thus, there exist $a, b \in \mathbb{Z}$ s.t.

$$\zeta^a - 1 = \underbrace{la + lb \cdot \zeta}$$

$(1, \zeta)$ a basis as $\mathbb{Z}$-Modul. and $a \neq 1$

But this is absurd. → false. $\mathbb{Z}_K$ monogenic doesn't mean it has rank 2.

If $\zeta^a - 1 \in \underbrace{Q/l\mathbb{Z}_K}$.: → quotient?!

It holds that

$$\min_{K/\mathbb{Q}}(\zeta) = \prod_{i=1}^{p-1} (1 - \zeta^i)$$

But Th since every embedding $\sigma_1, \dots, \sigma_{p-1}$ mapps a root of min to another root, we get that

$$N_{K/\mathbb{Q}}(\zeta^a - 1) = p.$$

But by the multiplicativity of the norm we get $N((\zeta - 1)\mathbb{Z}_K) = p$. But since every prime ideal div. divides l, we get a contradiction since p, l are coprime. ☒

Nuclear final part

1.5